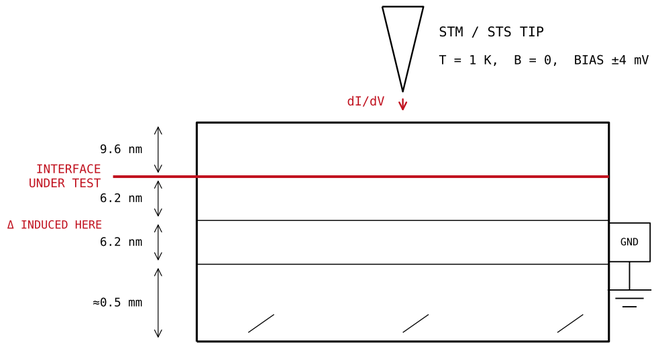
MILESTONE 1 — PROXIMITY-INDUCED GAP AT A TWISTED-CUPRATE / TOPOLOGICAL-INSULATOR INTERFACE

sample definition, measurement, and decision criterion — one experiment, one number

1. TEST SAMPLE — SECTION

vertical NTS; no gates, no channel, no applied field

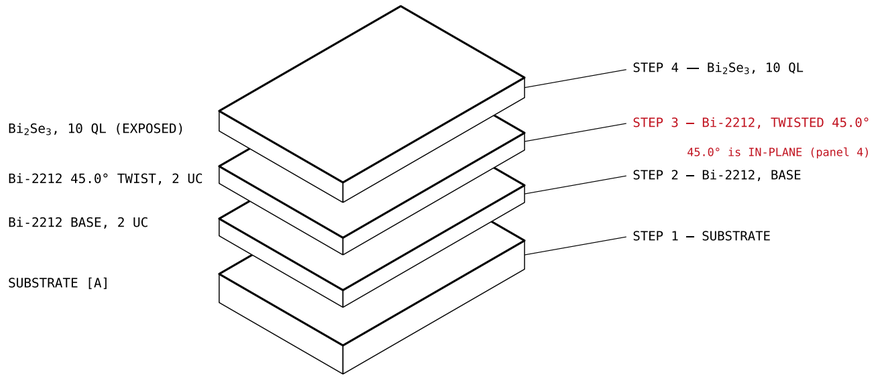


STACK BILAYER FIRST, TI LAST → PROXIMITISED SURFACE IS THE ONE THE TIP SEES

TRANSFER THE Bi_2Se_3 — DO NOT GROW IT ON THE CUPRATE (see panel 5)

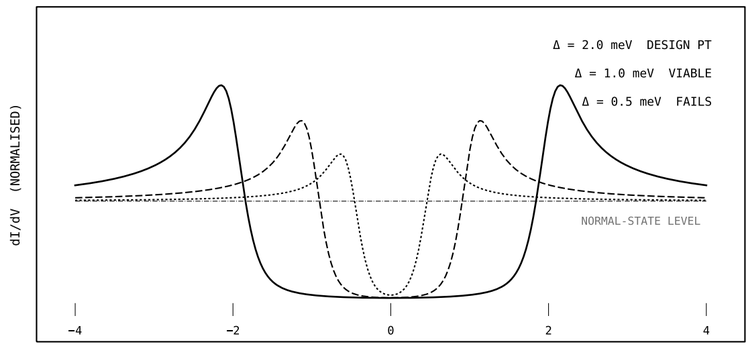
2. ASSEMBLY ORDER — EXPLODED ISOMETRIC

opaque, hidden lines removed; NTS; assembled sample has zero gaps



3. PREDICTED SPECTRA — COMPUTED [M]

Dynes-broadened, Γ = 0.05Δ, thermally smeared at T = 1 K

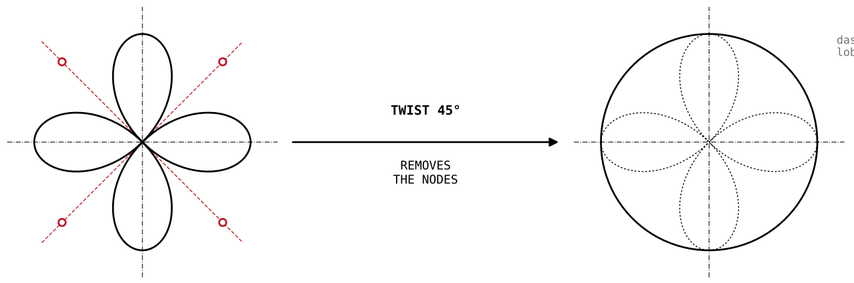


OBSERVABLE: HARD GAP — ZERO-BIAS dI/dV SUPPRESSED TO A FEW % OF NORMAL

A SOFT GAP (≥0.3 OF NORMAL) IS ITSELF AN INFORMATIVE RESULT

4. WHY THIS CONFIGURATION — GAP ANISOTROPY |Δ(θ)| ON THE FERMI SURFACE

the published null result was measured on the LEFT-HAND case; the right-hand case has not been tested



(i) SINGLE Bi-2212 LAYER

$d_{x^2-y^2}$: $|\Delta| \propto |\cos 2\theta|$

FOUR NODES — $|\Delta| = 0$ ALONG (110)

(ii) 45° TWISTED BILAYER

$d + i\,d'$: $|\Delta| = \text{CONST}$

NO NODES — $|\Delta| > 0$ IN EVERY DIRECTION

5. PRIOR ART ON THIS INTERFACE — AND WHAT IS UNTESTED

E. WANG et al., arXiv:1403.4184 (2014)

Bi_2Se_3 grown in situ on Bi-2212. NO induced gap observed.

SINGLE LAYER → NODAL. Symmetry argument predicts failure.

P. WANG et al., Nat. Commun. 3, 1056 (2012)

Mechanically bonded Bi_2Se_3/Bi-2212. Proximity gap reported to 80 K.

SINGLE LAYER, but bonded rather than grown. Positive.

RACHMILOWITZ et al., npj Quantum Mater. 5, 72 (2020)

Bi_2Te_3 on Bi-2212. V-shaped in films → HARD GAP on islands.

COULOMB BLOCKADE, not pairing. A POSITIVE RESULT MUST EXCLUDE IT.

MICROSOFT QUANTUM, arXiv:2606.03884 (2026)

InAs-Pb tetron: Δ_{ind} 570 μeV → Δ_T 70 μeV, measured at 50 mK.

BENCHMARK: 12% induced-to-topological retention in real hardware.

THE 45° TWISTED CONFIGURATION HAS NOT BEEN TESTED.

The twist converts the pairing to fully gapped d+id′, so the nodal symmetry objection that explains the 2014 null result does not apply to it.

6. DECISION CRITERION — WHAT THE MEASURED NUMBER DECIDES

MEASURED Δ	→ Δ_{top} [M]	T_{max} [M]	VERDICT
< 0.50 meV	< 0.52 meV	< 0.30 K	ARCHITECTURE FAILS
1.00 meV	1.002 meV	0.58 K	VIABLE
2.00 meV	1.944 meV	1.13 K	DESIGN POINT
3.00 meV	2.825 meV	1.64 K	COMFORTABLE

EVERY DOWNSTREAM NUMBER IN THE ARCHITECTURE SCALES FROM THIS ONE MEASUREMENT.

Δ_{top}/Δ ABOVE IS THE CLEAN LIMIT (0.97); REAL DEVICES REACH 0.12 (panel 5). WE CROSS THE THRESHOLD NEAR $\delta_{rms} \approx \Delta$ — MANUSCRIPT SEC. VI.

7. WHAT WOULD MAKE A RESULT CONVINCING

CONTROL 1 — TEMPERATURE. A proximity gap closes at the cuprate T_c (~90 K); a Coulomb-blockade gap does not. dI/dV vs T is the discriminator.

CONTROL 2 — UNTWISTED REFERENCE. A 0° stack from the same run should show the nodal V-shape of panel 4(i); the 45° stack a hard gap.

CONTROL 3 — ISLAND SIZE. A charging gap scales with inverse capacitance and varies with lateral extent; a proximity gap does not (ref. above).

SAMPLE. Ground contact to the bilayer required; tens of μm lateral extent [A].

OPEN — FOR YOUR VIEW. Angular tolerance on the twist is not computed, and whether cryogenic two-cuprate assembly transfers to a cuprate/TI interface cannot be settled by simulation.

A NULL RESULT DECIDES THE ARCHITECTURE AS FIRMLY AS A POSITIVE ONE.

NOTES:

1. PROVENANCE: [M] COMPUTED FROM THE VALIDATED BdG MODEL (tvqpu.lattice, 17-CHECK GATE); [L] LITERATURE; [A] ASSUMED.
2. ALL LAYER THICKNESSES ARE [A]. THE MEASUREMENT REQUIRES ONLY A PROXIMITISED TI SURFACE ACCESSIBLE TO A PROBE.
3. PANEL 3 USES THE SAME MODEL AS THE DECISION TABLE.
4. SECTION AND ISOMETRIC ARE VERTICALLY NTS.
5. GRADY 2026 — github.com/VariableYT/Twisted-Cuprate-Majorana

VS

VARIABLE SYSTEMS

EL DORADO HILLS, CA

GEOMETRY SHEET — NOT FABRICATION DATA

PROJECT:	TWISTED-CUPRATE MAJORANA CHANNEL — MILESTONE 1
SUBJECT:	PROXIMITY GAP AT Bi-2212(45°) / Bi_2Se_3 INTERFACE
DRAWING:	SECTION / ISO / SPECTRA / ANISOTROPY / DECISION

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